



Summary of Corrections and Enhancements

Delivery Id: OR19 Server PP9.3
Product Family: Platform Products
Product: Nokia AirFrame Open Rack
Release: OR19 Server eSW

Approval date: 26-Jan-2024

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Table of Contents

- 1. Purpose.....4
- 2. Definitions and Abbreviations5
- 3. Package Identification Details6
 - 3.1 eSW binaries6
 - 3.2 eSW tools6
 - 3.3 Peripherals ISO image6
 - 3.4 Open Source SW8
- 4. Fault Corrections.....9
 - 4.1 Security vulnerability corrections.....9
 - 4.2 Non-security corrections.....9
- 5. New Features and Changes10
 - 5.1 New features10
 - 5.2 Other SW changes10
- 6. 3rd Party Corrections.....11
 - 6.1 3rd party security vulnerability corrections.....11
- 7. List of Known Issues.....12
 - 7.1 Embedded software known issues.....12
 - 7.2 Security issues12
 - 7.3 Restrictions and limitations12
- 8. Previous Deliveries for the Product.....14
- 9. Compatibility Details.....15
 - 9.1 HW revision compatibility15
 - 9.2 SW revision compatibility.....15
 - 9.3 Interface compatibility.....15
- 10. Appendices / References16

Contact

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Summary of changes:

Date	Version	Changes
26-Jan-2024	1.0	Approved version.

1. PURPOSE

The purpose of this document is to give a summary of the corrections and enhancements for Priority Package 9.3 (PP9.3) of Nokia AirFrame Open Rack release OR19 Server eSW. It also includes content of current known issues, restrictions and limitations, and compatibility information from Server eSW, commodity FW update utility (ISO) and Server Diagnostic Tool. It does not include known issues, restrictions and limitations, and compatibility information from commodity FW itself. Those are usually publicly available on commodity vendor's website.

This Maintenance Package is valid also for OR18 Server eSW release.

NOTE:

Priority Package Release is an incremental package generated over the previous Feature/Maintenance Package Release. Priority Package Release must be installed over the previous Feature Package/Maintenance Package Release.

2. DEFINITIONS AND ABBREVIATIONS

BIOS	Basic Input Output System
BMC	Board Management Controller
ESW	Embedded Software
FW	Firmware
HW	Hardware
NADCM	Nokia AirFrame Data Center Manager
NDCS	Nokia Data Center Solution
OR	Open Rack
WEBUI	Web User Interface

3. PACKAGE IDENTIFICATION DETAILS

3.1 eSW binaries

Category	HW Unit	Component / Model	Version	eSW Package / Image Name
Server	AO-SER2U-AS AO-SER2U-BS	BIOS FW	4.21.10	BIOS_NDCS18_Package_4.21.10.tar.gz
		BMC FW	3.65.00	BMC_NDCS18_Package_3.65.00.tar.gz
		CPLD FW	0.1.9	
		MIB file	1.02	NSN-OPENRACK-MIB_1.02.txt
		Peripheral FW ISO Image (Refer to Chapter 3.3 for detailed FW version info)	3.8.0	NDCS_PERIPHERALS_FW_UPDATE_ISO_V3.8.0.tgz

3.2 eSW tools

Category	HW Unit	Component / Model	Version	eSW Package / Image Name
Server	All servers	Yafuflash FW Update Tool	5.8.0	AO-SER_Yafuflash_5.8.0.tgz
		BMC Info Collection Tool	01.00.00	AO-SER_eSWTool02_01.00.00.tgz

3.3 Peripherals ISO image

The ISO image is applicable for servers which have specified peripheral devices equipped. The supported peripheral device models' and target FW versions' info as below:

HW Unit	Name / Model	Product Number	FW Version
RAID Controller	MegaRAID-9380-8E	LSI00438	FW: 24.21.0-0159
HBA Card	SAS9300-8e	LSI00343	FW: 16.00.01.00 BIOS: 08.37.00.00 UEFI BSD: 18.00.00.00
NIC Card	XXV710DA2	XXV710DA2G1P5	NVM: 9.10 OROM: 27.8
	XXV710DA2 OCP	XXV710DA2OCP1	
	ConnectX-4 Lx EN MCX4421A-ACAN (OCP)	MT_2470111034	FW: 14.32.1010 PXE: 3.6.0502 UEFI: 14.25.0017
	ConnectX-4 Lx EN MCX4121A-ACAT	MT_2420110034	
	ConnectX-4 Lx EN MCX4421A-ACQN (OCP)	MT_2470112034	
	ConnectX-5 Ex EN MCX516A-CCAT	MT_0000000012	FW: 16.35.2000 PXE: 3.6.0805 UEFI: 14.28.0016
	ConnectX-5 VPI MCX545A-ECAN (OCP)	MT_0000000077	
	ConnectX-5 EN MCX542A-ACAN	MT_0000000167	

HW Unit	Name / Model	Product Number	FW Version
	ConnectX-5 EN MCX512F-ACAT (OCP)	MT_0000000183	
SSD	S4600	SSDSC2KG480G701 SSDSC2KG960G701 SSDSC2KG019T701	FW: SCV10150
	S4510	SSDSC2KB960G801 SSDSC2KB019T801 SSDSC2KB038T801	FW: XCV10160
	S4610	SSDSC2KG480G801 SSDSC2KG960G801 SSDSC2KG019T801 SSDSC2KG038T801	
	S4510 (SATA M.2)	SSDSCKKB480G801	FW: XC311151
	S4520	SSDSC2KB960GZ01 SSDSC2KB019TZ01 SSDSC2KB038TZ01	FW: 7CV10120
	S4620	SSDSC2KG480GZ01 SSDSC2KG960GZ01 SSDSC2KG019TZ01 SSDSC2KG038TZ01	
	S4520 (SATA M.2)	SSDSCKKB480GZ01	
	P4600	SSDPE2KE016T701 SSDPE2KE032T701	FW: QDV101D1
	P4610	SSDPE2KE016T801 SSDPE2KE032T801	FW: VDV10184
	5100MAX	MTFDDAK480TCC-1AR1ZABYY	FW: DOMU075
	5100PRO	MTFDDAV480TCB-1AR1ZABYY	
	5200MAX	MTFDDAK480TDN-1AT1ZABYY MTFDDAK960TDN-1AT1ZABYY MTFDDAK1T9TDN-1AT1ZABYY	FW: D1MU030
	9200MAX	MTFDHAL1T6TCU-1AR1ZABYY MTFDHAL3T2TCU-1AR1ZABYY	FW: 101008S0
	9300MAX	MTFDHAL3T2TDR-1AT1ZABYY	FW: 11300DY0
	SM863a	MZ7KM480HMHQ-00005 MZ7KM960HMJJP-00005 MZ7KM1T9HMJJP-00005	FW: GXM5304Q
	SM883	MZ7KH480HAHQ-00005 MZ7KH960HAJR-00005 MZ7KH1T9HAJR-00005 MZ7KH3T8HALS-00005	FW: HXM7A04Q
	PM883	MZ7LH960HAJR-00005 MZ7LH1T9HMLT-00005 MZ7LH3T8HMLT-00005	FW: HXT7B04Q
	PM893	MZ7L3960HCJR-00A07 MZ7L31T9HBLT-00A07 MZ7L33T8HBLT-00A07	FW: JXTC304Q

HW Unit	Name / Model	Product Number	FW Version
SSD	PM897	MZ7L3480HBLT-00A07 MZ7L3960HBLT-00A07 MZ7L31T9HBNA-00A07 MZ7L33T8HBNA-00A07	FW: JXTE204Q
	SN840	WUS4C6416DSP3X1 WUS4C6432DSP3X1	FW: R2210010
HDD	Exos 7E8	ST6000NM0095	FW: E004
	MG06	MG06SCA10TE	FW: 0108
	MG07	MG07SCA12TE	FW: 0107
JBOD	AO-JBOD45-A	AO-JBOD45-A	Expander: 01.04.01.20 MFG: 00.14 CPLD: 10
FC Card	LPe32002	LPe32002-M2	FW: 12.2.299.14

3.4 Open Source SW

Category	HW Unit	Component / Model	SW Package / Image Name
Open Source	All servers	Open Source code resources	OR19_Server_PP9.3_OpenSource.zip

4. FAULT CORRECTIONS

4.1 Security vulnerability corrections

Nokia Fault ID	Description
NA	NA.

4.2 Non-security corrections

Nokia Fault ID	PR730659 [Major]
Fault Description	BMC can't get default IPv6 route from RA when IPv6 changes from DHCP to static.
Root Cause Analysis	BMC doesn't accept RA when IPv6 changes to static mode.
Correction Description	[Correction in Server BMC 3.65.00] Accept RA when IPv6 changes to static mode.

Nokia Fault ID	PR745565 [Major]
Fault Description	OR18 servers BMC can't be accessed via IPMI, Redfish, BMC WEBUI and serial interface.
Root Cause Analysis	Some BMC processes don't delete BMC logs properly and completely.
Correction Description	[Correction in Server BMC 3.65.00] Delete BMC logs properly to avoid some logs are kept forever.

Nokia Fault ID	PR745599 [Minor]
Fault Description	Lan port info causes NADCM confusion when changing IPv6 from DHCP to static.
Root Cause Analysis	BMC does not check address mode when dynamic IPv6 address is requested.
Correction Description	[Correction in Server BMC 3.65.00] Set IPv6 address to 0 when dynamic IPv6 address is requested under static mode.



5. NEW FEATURES AND CHANGES

5.1 New features

Package Name / Version	Description
NA	NA.

5.2 Other SW changes

Package Name / Version	Description
Server ISO 3.8.0	Add new FW support: - SSD model S4620 SSDSC2KG038TZ01 - HDD model MG06 MG06SCA10TE - HDD model MG07 MG07SCA12TE
	Update scripts to support NVMe FW encrypted update.
	Change auto-shutdown action to force power off.



6. 3RD PARTY CORRECTIONS

6.1 3rd party security vulnerability corrections

Package Name / Version	Description
NA	NA.

7. LIST OF KNOWN ISSUES

7.1 Embedded software known issues

The detail content of embedded software known issues is not mentioned in this document for this PP delivery. Please refer to the latest MP delivery instead.

7.2 Security issues

The detail content of Security issues is not mentioned in this document for this PP delivery. Please refer to the latest MP delivery instead.

7.3 Restrictions and limitations

Package Name / Version	Description
Server BIOS 4.21.10	After BIOS upgrade, the first attempt to change BIOS parameters via Redfish may not take effect if the BIOS boot mode was LEGACY in the inactive flash bank before performing the BIOS upgrade. The user must double check if BIOS parameters are properly configured as expected. if not, the BIOS configuration must be repeated to fix the inconvenience.
Server BMC 3.65.00	When BMC firmware upgraded from v2.06.2 to v3.07.02 and later with "Configuration Preservation", user password will be requested to reset in WebUI. User can send an IPMI raw command after BMC was upgraded as a workaround (to disable resetting user password): \$ ipmitool -I lanplus -H {IP} -U {username} -P {password} raw 0x30 0x51 0x02 0x01 0x01
	When BMC firmware upgraded from early versions to v3.13.06 and later, Two BMC image blocks will be updated during BMC upgrade because of new uboot version.
	When BMC firmware upgraded from early versions to v3.21.20 and later, SNMP user authentication failed issue will be happened, and downgrade vice versa.
	When BMC firmware downgraded from v3.21.20 and later to early versions, all IPMI user login operation will be failed.
	When BMC firmware upgraded from early version to 3.42.00 and later, redfish configuration cannot be preserved and vice versa.
Server ISO 3.8.0	Manual Power Cycle on JBOD is needed for new CPLD FW Activation.
	Not support FW update for JBOD drives in UBad (Unconfigured Bad) or Offln (Offline) state.
	The update tool ONLY supports FW Upgrade to later versions for devices (components), which means no update action if the current version is already up to date or later than the target version.
Server Diagnostic Tool 01.13.00	After HW diagnostic, manual power on the payload since the payload will be power off after diagnostic test cases completed.
	If there is huge "Uncorrectable ECC" errors with DIMM memory, the diagnostic tool will not be able to complete the whole diagnostics procedure normally. To solve the issue, check SEL log to locate the faulty memory chip and replace it first, and then execute the diagnostic tool.



Package Name / Version	Description
	No input or action is allowed while Diagnostic tool is running. Please don't press any keyboard while Diagnostic tool is running.

8. PREVIOUS DELIVERIES FOR THE PRODUCT

History information for the same product deliveries. In case of generic SW, this table does not include customer specific deliveries.

Date	Delivery Name
20-Sep-2018	OR18 Server
17-Dec-2018	OR19 Server FP2
03-Apr-2019	OR19 Server FP4
31-Jul-2019	OR19 Server FP5
04-Dec-2019	OR19 Server PP5.1
06-Nov-2019	OR19 Server FP6
28-Feb-2020	OR19 Server PP6.1
20-May-2020	OR19 Server PP6.2
07-Jul-2020	OR19 Server PP6.3
15-Jul-2020	OR19 Server FP7
18-Dec-2020	OR19 Server PP7.1
12-Mar-2021	OR19 Server MP8
12-Jan-2022	OR19 Server PP8.1
07-Sep-2022	OR19 Server MP9
05-Jan-2023	OR19 Server PP9.1
22-Aug-2023	OR19 Server PP9.2

9. COMPATIBILITY DETAILS

9.1 HW revision compatibility

HW Product	HW Revision	Supported eSW Versions
AO-SER2U-AS AO-SER2U-BS	.01 and later revisions	BMC 2.06.2 and later versions BIOS 2.05.10 and later versions CPLD 0.1.9 and later versions ISO Image 2.5 and later versions

9.2 SW revision compatibility

Delivery ID	NADCM compatible with the Delivery
OR19 Server PP9.3	24R1-NADCM and later NADCM deliveries.

9.3 Interface compatibility

Delivery ID	Redfish API compatible with the Delivery
OR19 Server PP9.3	Specification Version: 1.7.0 (DSP0266) Schema Version: 2019.1 (DSP8010)

Delivery ID	SSH compatible with the Delivery
OR19 Server PP9.3	User need to choose “ Don’t preserve Authentication configuration ” to upgrade BMC image to/downgrade BMC image from version 3.13.06, as default user name and password for SSH is changed in BMC 3.13.06 onwards. Otherwise, user would not create SSH session.

10. APPENDICES / REFERENCES

Name	Nokia AirFrame Data Center Solution, NDCS OR 19, Operating Documentation AirFrame Data Center Solution, Rel. NDCS OR 19 PP9.2, Issue 12
Link	https://customer.nokia.com/support/ https://doc.networks.nokia.com/product/Nokia_AirFrame_Data_Center_Solution/134-080698.00/release/NDCS_OR_19/833-045659.00/collection/NDCSOr/0
Remarks	This is the reference document set for AirFrame HW and NDCS embedded SW. Logging in to Nokia Support portal or Discovery Center is required first to make the direct link to the Operating Documentation work.

Name	Nokia AirFrame Data Center Solution, NDCS OR 18, Operating Documentation AirFrame Data Center Solution, Rel. NDCS OR 18, Issue 03
Link	https://customer.nokia.com/support/ https://doc.networks.nokia.com/product/Nokia_AirFrame_Data_Center_Solution/134-080698.00/release/NDCS_OR_18/833-005459.00/collection/NDCSOr/0
Remarks	This is the reference document set for AirFrame HW and NDCS embedded SW. Logging in to Nokia Support portal or Discovery Center is required first to make the direct link to the Operating Documentation work.